INCH/POUND
MIL-A-36188B
30 June 1989
SUPERSEDING
MIL-A-36188A
6 August 1974

MILITARY SPECIFICATION

APRON, DENTAL, PLASTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a plastic apron suitable for use as a protector from debris, tissue exudates and water used in dental operative procedures.

2. APPLICABLE DOCUMENTS

2.1 GOVERNMENT DOCUMENTS

2.1.1 Specifications and standards. The following specifications standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6532
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SPECIFICATIONS

Federal

L-P-375	- Plastic Film, Flexible, Vinyl Chloride.
V-T-276	- Thread, Cotton.
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, Bleached, or Dyed).
PPP-B-566	- Boxes, Folding, Paperboard - Less Than 20 pounds.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Locked-covered.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Boxes, Setup.

Military

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.
MIL-F-21840	- Fastener, Tapes, Hook and Pile, Synthetic.

STANDARDS

Federal

FED-STD-406	- Plastic, Methods of Testing.
FED-STD-751	- Stitch, Seam, and Stitching.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Naval Publications Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

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DRAWING

Defense Personnel Support Center (DPSC)

No. 24347 - APRON, DENTAL, PLASTIC

(Copies of drawing may be obtained from the Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (execpt for related associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Vinyl plastic. The plastic sheeting shall be a flexible vinyl waterproof film conforming to type I of L-P-375, milky white or opaque in color, with a smooth finish, except that the following characteristics shall not apply: ultimate elongation, resistance to weathering, cold crack, and lacquer lifting. Blocking shall be determined as specified in method 1131 of FED-STD-406 at 70 plus or minus 1 degrees centigrade (°C).

3.1.1.1 Apron Body. The apron body sheeting shall be 4 gage (0.004 inch), 5 gage (0.005 inch) or 6 gage (0.006 inch), plus or minus 10 percent per gage, plastic sheeting (see 3.1.1).

3.1.1.2 Neckband. The neckband sheeting which forms the collar shall be 12 gage (0.012 inch) to 16 gage (0.016 inch), plus or minus 10 percent per gage, plastic sheeting (see 3.1.1).

3.1.2 Nylon tape fastener. The nylon tape fastener shall be white, type I, class 1 hook and pile in accordance with MIL-F-21840.

3.1.3 Cotton.

3.1.3.1 Cotton binding. The cotton binding used for finishing the edges of apron, except for neck closure, shall be preshrunk, white cotton tape, type I, or type III, class 2, of DDD-T-86. As an alternate, for binding the edges of apron, bleached cotton sheeting of 100/60 designation, conforming to a good commercial quality, may be used.

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3.1.3.2 Thread. The thread shall be white, 3 ply cotton, type IA2 ticket No. 40 or 50, or type IC1, ticket No. A or O, in accordance with V-T-276.

3.2 Design and construction. The apron shall be suitable for patient or operator use during dental operative procedures. The plastic apron shall consist essentially of an apron body having all edges, except neck hole opening, reinforced by machine sewn cotton binding; two tie tapes at the waistline; a centrally located keyhole type neck opening; reinforced with 2-ply vinyl plastic sheeting (see 3.2.1) forming a collar, with an adjustable neck closure to accommodate a range of neck sizes.

3.2.1 Apron body. The apron body shall be made of one-piece flexible vinyl plastic film as specified in 3.1.1 and 3.1.1.1. The apron body shall conform to the design and dimensions shown in Defense Personnel Support Center (DPSC) Drawing 24347.

3.2.2 Cotton binding. All edges of the apron body, with the exception of the neck line, shall be reinforced by white, bleached cotton tape or sheeting as specified in 3.1.3.1. The cotton binding shall terminate at the ends of neckband and shall be folded over on itself and be bartacked by a minimum of 42 stitches. The cotton binding, made of one (1) inch wide stock tape or sheeting, shall be machine sewn with white cotton thread (see 3.1.3.2), using a 301 or 401 class stitch in a BSc-1 type seam, in accordance with FED-STD-751. The seam shall have 8 to 12 stitches per inch. The dimensions of finished binding shall be as shown in DPSC Drawing 24347.

3.2.3 Ties. A tie shall be securely fastened to each side of the apron body by means of a 42 stitch bartack, using cotton binding specified in 3.1.3.1. The cotton binding shall be machine sewn as described in 3.2.2, except that a SSe2b or EFp-1 type seam shall be used. The free ends of the tie tapes shall be finished to prevent ravelling. The dimensions of the ties shall be as shown in DPSC Drawing 24347.

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3.2.4 Neckband. The neck opening, centered in the top end of the apron body shall be finished with a 2-ply plastic material as specified in 3.1.1.2, forming a stand-up, open, circular collar. The 2-plies shall be machine sewn together using white cotton thread conforming to 3.1.3.2, with a 401 class stitch in an SSe-2b or SSc-1 type seam for the upper neckband edge and an SSq-2b or LSe-1 type seam for the lower neckband edge, in which the apron body neckline material shall be included as the middle ply. Stitches and seams shall be in accordance with FED-STD-751. The finished, attached neckband shall conform to the dimensions shown in DPSC Drawing 24347. The neckband shall have a hook and pile adjustable nylon fastener tape closure as specified in 3.1.2 The hook shall be machine sewn onto the left outside face of the neckband. The pile shall be machine sewn onto the right inside face of the band. The hook and pile shall be machine sewn using cotton thread conforming to 3.1.3.2, with 14 to 16 class 301 or 401 stitches per inch in accordance with FED-STD-751. The dimensions and positioning of the hook and pile shall be as shown in DPSC Drawing 24347.

3.3 Laundering. The finished apron shall be suitable for laundering with lukewarm water and mild soap when tested as specified in 4.4.1. The apron shall remain flexible, soft, colorless, and intact, showing no signs of stiffening, shredding, tearing or disintegrating. The marking on the identification label shall remain legible and permanent after laundering.

3.4 Identification marking. A label shall be sewn into the bottom left corner of the binding of the apron. The label shall be permanently marked with the manufacturer's name or registered trademark.

3.5 Workmanship. The apron shall be free of missing components, imbedded material, cuts, tears, holes and any defect which detracts from its appearance or impairs its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit to the Government to accept defective material.

4.1.2 Certificates of compliance. Certificates of compliance supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.2.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.3 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, III and IV (see 4.3.1, 4.4.1, 4.4.2, and 4.4.3) and table II (see 4.3.2).

4.3.1 Sampling for examination of apron defects. Sampling for visual and dimensional examination (see 4.3) shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as specified in table I. Unit of product for examination shall be one apron when the examination is performed on the end item; unit of product for examination shall be the applicable component part of the apron when the examination is performed on a component part basis.

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TABLE I. Sampling for examination.

	Inspection level	AQL'S Major	AQL'S Total
For defects applicable to 4.4.1	I	4.0	10.0

4.3.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.The inspection levels and AQLs shall be as indicated in table II. Unit of product for tests performed on end item shall be one apron (unit of issue). When tests are performed on raw material, a sufficient amount of each lot of raw material used shall be taken as the sample unit of product.

TABLE II. Tests.

Characteristics	Requirements	Test Method	No. of determinations per sample unit	Results reported as	Insp level	AQL
Dimensions	3.2.1 3.2.2 3.2.3 3.2.4	———	———	Pass or Fail	S-1	6.5
Laundering	3.3	4.5.1	1	Pass or Fail	S-1	1.0

4.4 Examination of apron defects. The defects found during examination shall be classified in accordance with 4.4.1 and 4.4.2.

4.4.1 Examination of apron, dental, plastic for visual characteristics.

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TABLE III. Classification of apron defects.

EXAMINE	DEFECTS	MAJOR	MINOR
Quality and finish or plastic material	Material other than specified	X	
	Material not color specified	X	
	Bubble, sharp crease, wrinkle blister, gouge or other imperfection		X
	Cut, tear, hole, break or crack	X	
	Finish not as specified		X
	Objectionable odor (other than odor typical to Vinyl Chloride		X
Design	Any characteristic not in accordance with the specified requirements	X	
Construction and Workmanship			
General	Apron body not one-piece construction	X	
	Component missing	X	
	Required operation omitted or not properly performed	X	
	Foreign matter or material imbedded in the plastic sheeting		X
	Seam misplaced, or out of alignment, by more than 1/4 inch	X	
	Any component or part misaligned malformed, or damaged	X	

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Table III. Classification of apron defects Continued.

EXAMINE	DEFECTS	MAJOR	MINOR
Seams and stitching	Seam type not as specified	X	
	Stitch type not as specified	X	
	More than maximum number of stitches per inch damaging material	X	
	More than maximum number of stitches per inch but not damaging material		X
	One stitch per inch less than specified minimum		X
	Two or more stitches per inch less than the specified minimum	X	
	Any stitching omitted where required	X	
	Loose stitch and loosely secured seam		X
	Open seam-one or more stitches broken or 2 or more continuous stitches skipped on single stitch seam or on both rows of double stitch seam	X	
	Seam puckered or distorted		X
	Bartack missing, loose or misplaced	X	
Ties	Free end not properly finished	X	
	Soiled permanently		X
	Not secured adequately	X	

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Table III. Classification of apron defects Continued.

EXAMINE	DEFECTS	MAJOR	MINOR
Neckband	Not double thickness of plastic	X	
	Twisted in attachment to apron body so that it does not lie flat on neck	X	
	Ends of plastic band not completely bound by reinforcing cotton tape	X	
Fastener tape	Hook and pile not of type specified	X	
	Hook sewn on inside surface of neckband	X	
	Hook and pile do not function as specified	X	
	Hood and pile not positioned as required	X	
	Hook and pile not mounted as required	X	
	Hook and pile becomes detached during operation	X	
Identification marking	Omitted, incomplete, illegible, incorrect, or misplaced (marking defects scored once per apron regardless of number of faults)		X

4.4.2 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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Table IV. Classification of packaging defects.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Material	Any component missing
	Any component damaged, affecting serviceability
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling
	Bulging or distortion of containers
Weight or content (exterior and interior)	Number of intermediate packages or bundles is more or less than required gross/net weight exceeds requirements

4.5 Laundering Tests. The apron shall be immersed in a minimum of 2 gallons of one (1) percent neutral toilet soap solution maintained at 122 plus or minus 2 degrees fahrenheit (°F.) (50° plus or minus 1.1° C) for a period of one hour. The apron shall then be alternately dipped into and raised out of the soap solution for a period of 5 minutes, repeating the dipping cycle a minimum of 10 times per minute. This process shall be performed so as to assure exposure of entire apron surface to the soap solution. Upon removal after the test period, the sample shall be rinsed with warm water and shall be examined for characteristics specified in 3.3.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Apron shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each apron shall be neatly folded and inserted into a clear, snug fitting polyethylene bag having a minimum thickness of not less than 0.00125 inch. A 1/4 inch diameter hole shall be made at one corner of the bag to allow excess air to escape and the closure shall be effected by heat sealing or by means of a tuck or reverse flap.

5.1.1.4 Intermediate package. Twelve unit packages containing aprons shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for aprons shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Six intermediate packages (72 units) containing aprons shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Six intermediate packages (72 units) containing aprons shall be packed in and exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Six intermediate packages (72 units) containing aprons shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for apron shall be packed in accordance to the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in length, 52 inches in width, and 54 inches in height. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Apron packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2. Level C. Apron packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.2.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The plastic apron is intended to protect the dental patient from debris, tissue exudates and water used in dental operative procedures.

6.2.1. Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of specification.
(b) Selection of applicable levels of packaging and packing
(c) Applicable National Stock Number.

6.3 National Stock Number (NSN) Coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6532-00-890-1429	APRON, DENTAL, Plastic

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

6.5. Subject item (key word listing).

Apron
Binding, Cotton
Vinyl, plastic

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6532-1181

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-36188B	APRON, DENTAL, PLASTIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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